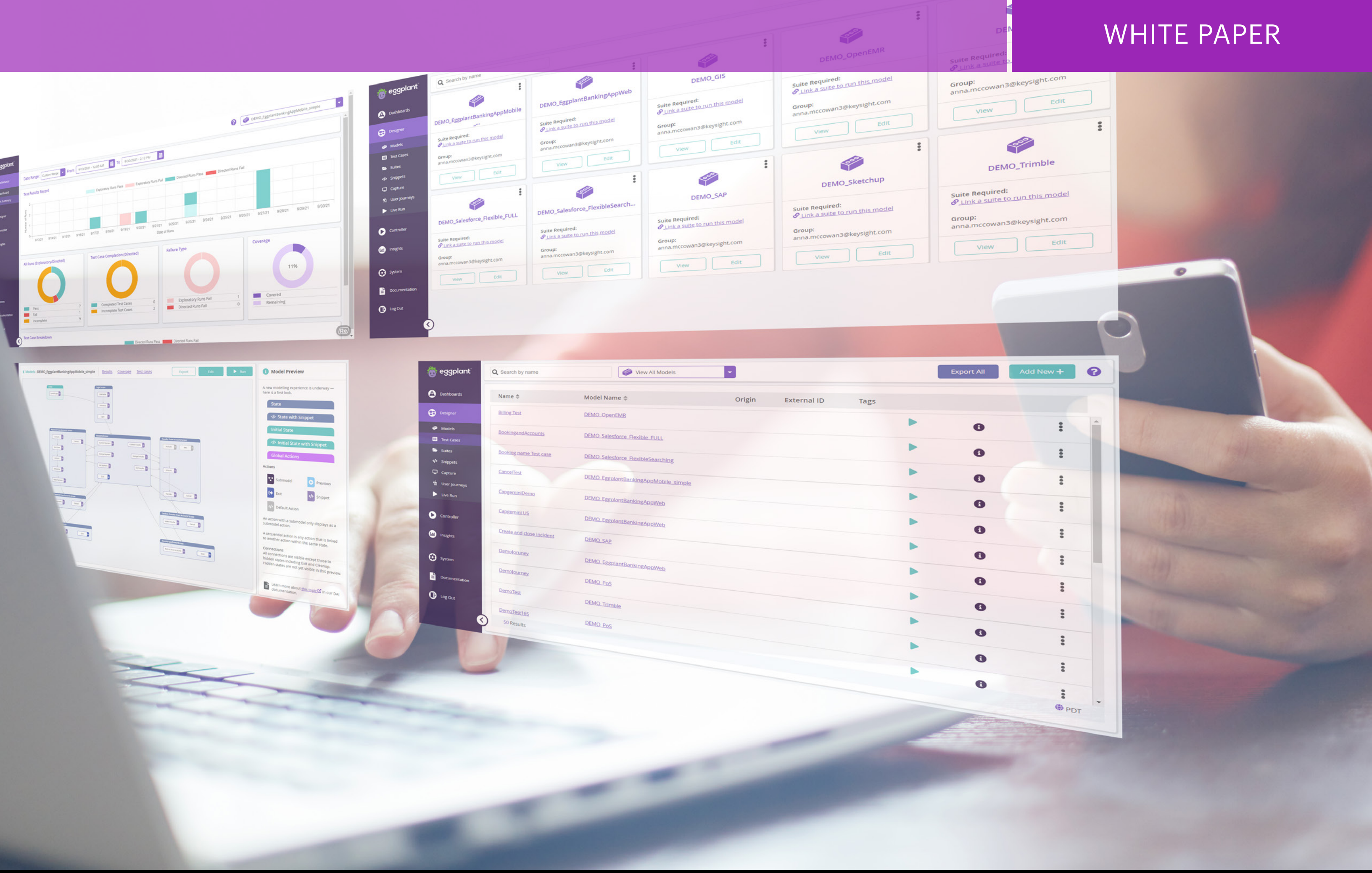
WHITE PAPER



Key Considerations for Continuous Test Automation

KEYSIGHT

eggplant
Test Automation Software

Key Considerations for Continuous Test Automation

Every company that relies on software is looking for new ways to increase the speed of delivery while also improving product quality. As a result, software development and quality assurance (QA) teams are under immense pressure to accelerate release cycles for each major update and new feature.

Modern software frameworks rely on microservices. Consequently, testing across the entire system has become demonstrably complex. A single application can have thousands of user journeys. When you multiply that by the number of browsers, devices, and operating systems, you quickly have a mountain of test cases that would take months to get through.

As a result, QA teams are often blamed for being the bottleneck in many organizations.

Backlogs of test cases need refactoring for regression testing, and testing must cover the impact of new code or features. Enhancements such as these increase the project's scope but with little to no added resources. Simply put, there's so much to test and not enough time.

Implementing continuous testing within the continuous integration and continuous delivery (CI / CD) pipeline helps QA teams quickly identify defects. QA can also notify project teams of the test status for prioritization and resolution.

What Is Continuous Test Automation?

Let's go over continuous test automation, what it is, and how you can align your test strategies with it.

Continuous test automation is a software testing methodology that uses automated tests at every point in the software delivery life cycle.

This process might involve some unit tests conducted after the developers have written code but before they push it or create a pull request.

After they make that pull request, some smoke tests may follow to ensure the build is stable. Then, there may be some more specific integration or feature tests of the application before merging regression testing.

Continuous test automation highlights defects early and gives developers time to fix them while the code is still fresh in their minds.

The main goal of continuous test automation is to ensure that the release candidate is constantly assessed for quality and improvement as it moves through the delivery pipeline.

Table 1. Benefits and challenges of continuous test automation

Benefits	Challenges
accelerating software deployment timelines	changing engrained mindsets between development and test teams
testing in tandem with agile development	ensuring test sets do not cause bottlenecks at stages of the software delivery pipeline
ensuring collaboration between QA and dev teams	ensuring a stable test environment to constantly run a high volume of tests
testing software as soon as it's built	
uncovering defects sooner	

A Common Focus on Quality

So what are some aspects to consider? First, continuous test automation allows testing to happen essentially in tandem with development. As soon as developers merge the code, QA teams can test it and provide quick feedback.

Continuous testing not only speeds up code delivery, but it also gives teams a shared focus on quality. It promotes greater communication between QA and development teams.

Additionally, companies are increasing the breadth of software that requires shipping — for example, mobile and desktop applications, cloud, and microservices. Testing these applications as soon as the development team builds them allows QA teams to highlight issues and initiate fixes sooner, without rushing at the end of a release cycle to get things fixed.

Shifting mindsets

As with any methodology change, there are challenges. Continuous test automation often requires a shift in mindsets across multiple teams.

Traditional testing involves relatively siloed teams whereby the developers mark an issue as merged or ready for testing, and then a separate QA team conducts the tests. It may require some cultural shifts to get the teams to work together or to empower the developers to perform a level of independent testing themselves. It's your code — make sure it works as designed.

Teams should also work together to consider which test packs to run and at which times. Additionally, consider how long these tests will take to run. A few smoke tests might be sufficient for most mergers, while a three-hour regression pack may cause bottlenecks in the pipeline.

It's important to note that if your team runs a high volume of tests, make sure you have a stable test environment that can handle the load.

Integrating Continuous Test Automation into Your Test Strategy

So how do you integrate continuous automation testing into your test strategy?

The most common way to do so is by using a continuous integration tool, such as Jenkins, Microsoft Azure, or Buddy. These all integrate seamlessly with Keysight Eggplant test automation software to initiate remote test runs and posttest results.

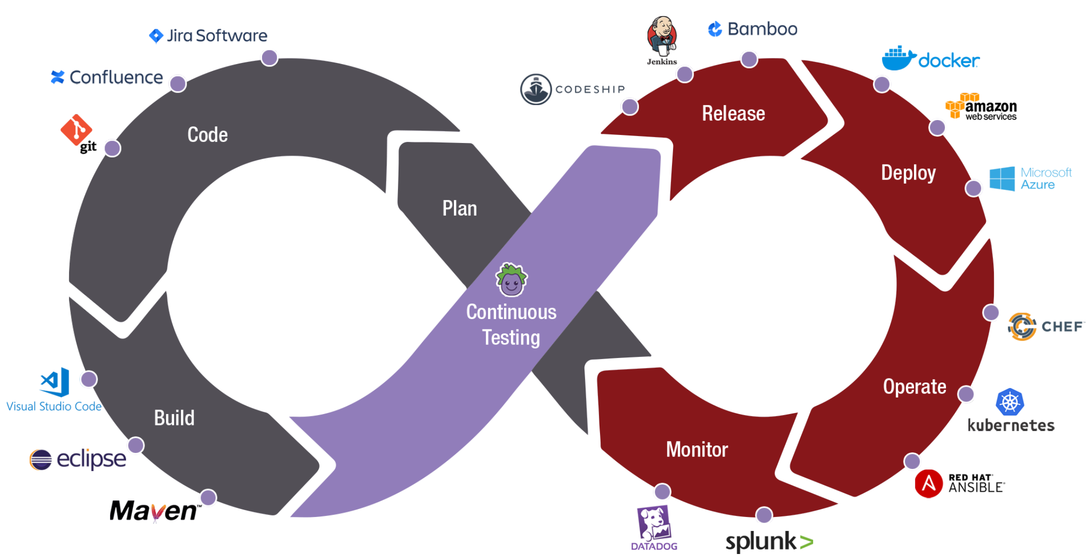


Figure 1. Eggplant Test and Automation Intelligence works with any existing technology to seamlessly integrate with tools essential for a CI/CD pipeline

For example, Jenkins can remotely run Eggplant test cases based on a trigger you specify beforehand. You can initiate your model test case or script run using your chosen parameters. Eggplant also integrates with test management tools such as Xray in Atlassian Jira, so you can instantly post results after each test run.

Eggplant accelerators provide connectivity with Jira to close the communication loop by raising a ticket or case ID. Developers can then prioritize issues in the next iteration. Keysight's Eggplant Test and Automation Intelligence can also automatically configure each test run with a Slack Webhook to post the test details to a chosen Slack channel.

About Eggplant

Keysight's Eggplant software is a low-code, easy-to-use test automation platform that seamlessly integrates continuous testing into DevOps build and release practices.

Eggplant easily integrates into agile DevOps processes and leading CI / CD tools to provide end-to-end testing across the full technology stack.

Learn more about how Eggplant can integrate with your continuous test automation strategy here.

Additional Resources

- Self-Guided Software Tour
- Keysight University course: Implementing Continuous Test Automation
- Solution Brief: Accelerate Software Delivery with Test Automation

For more information on Keysight Technologies' products, applications, or services, please visit: www.keysight.com